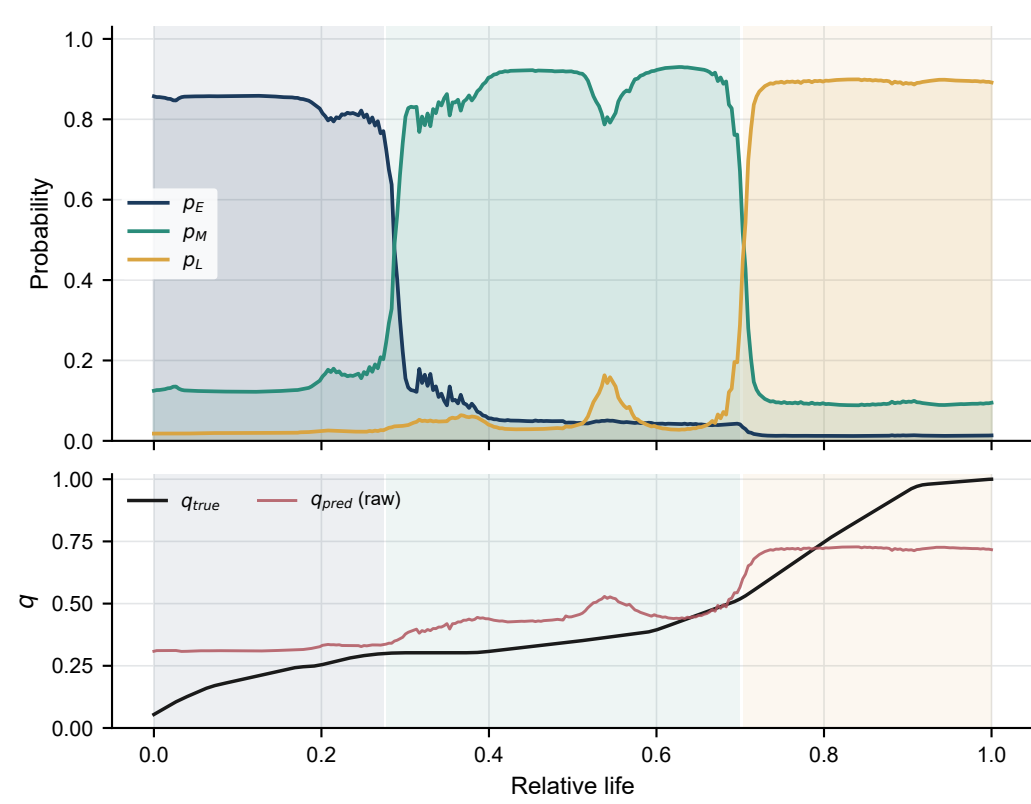
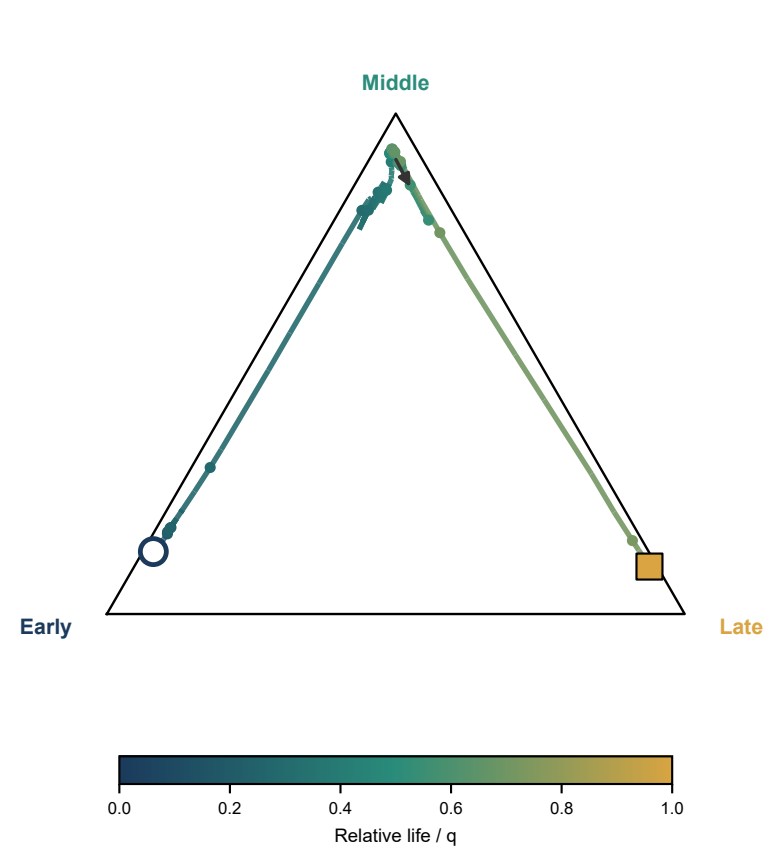
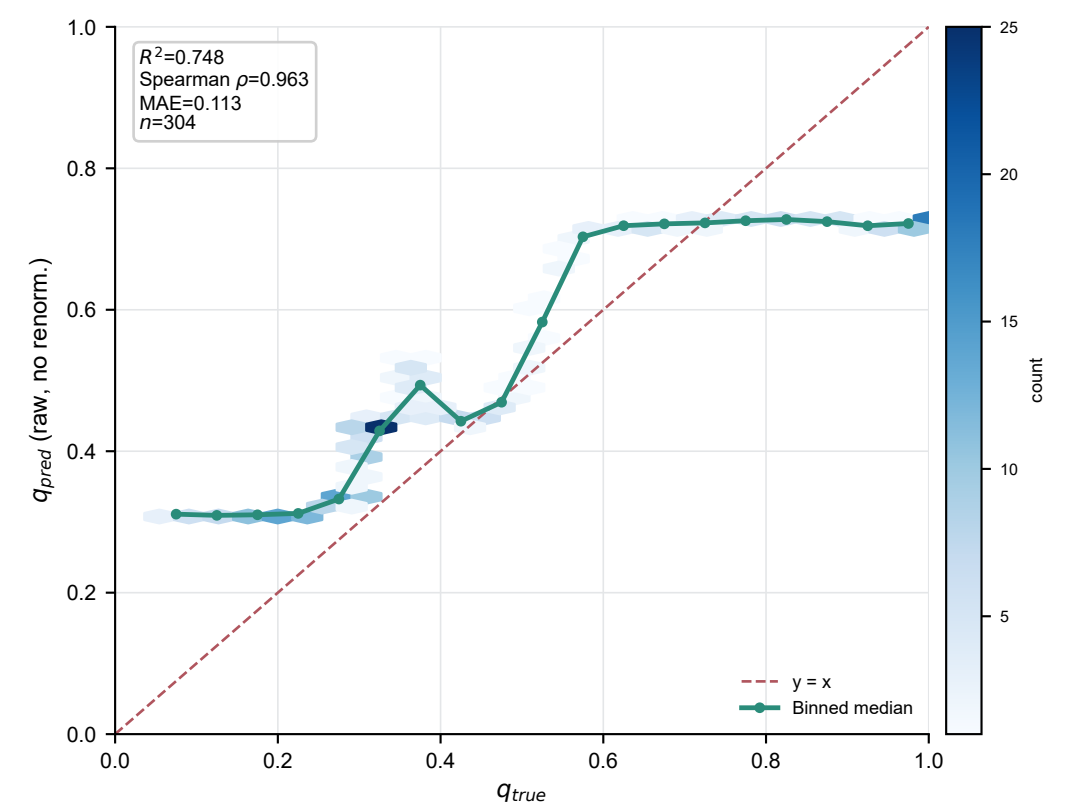




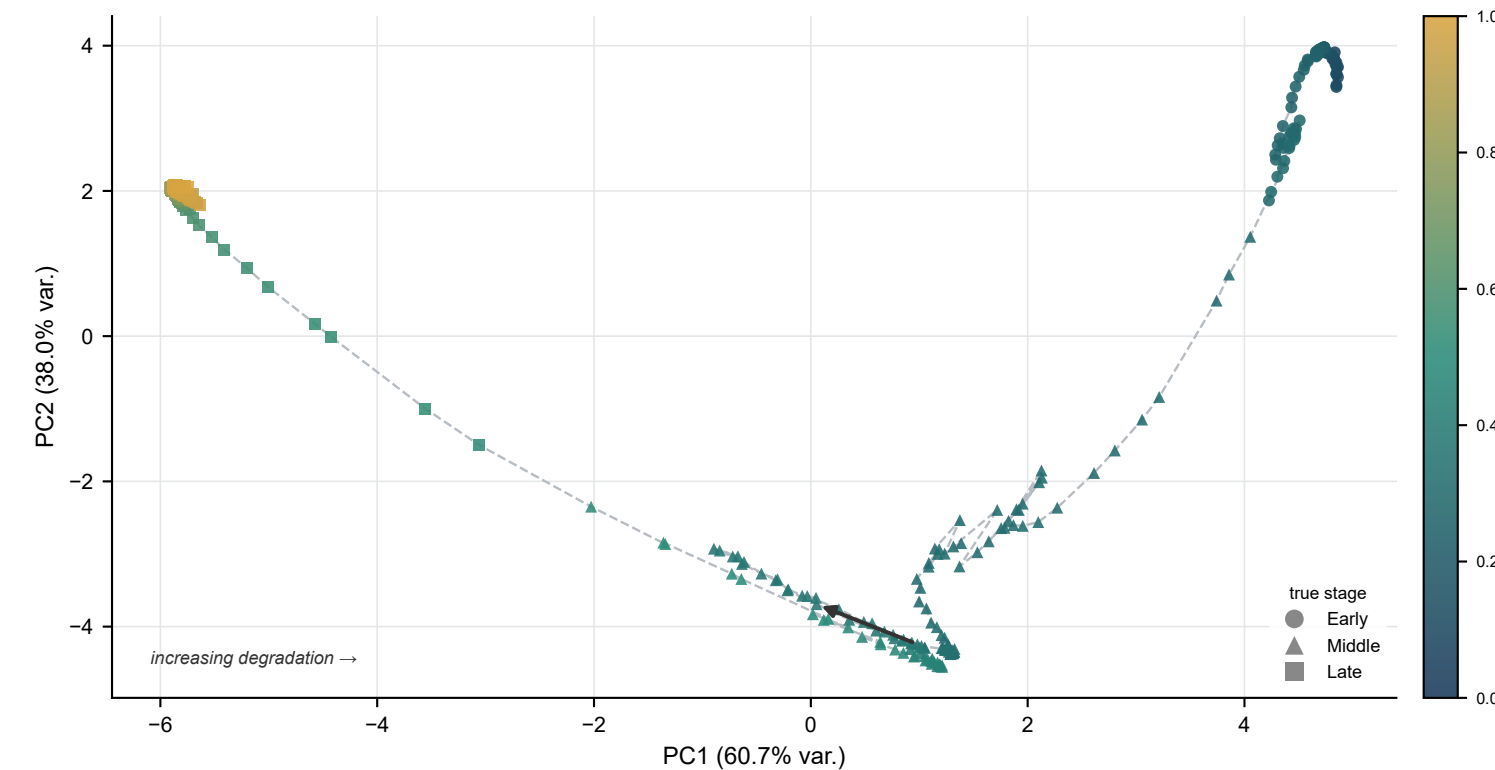
(a) Full C6 lifecycle: stage probability + degradation position



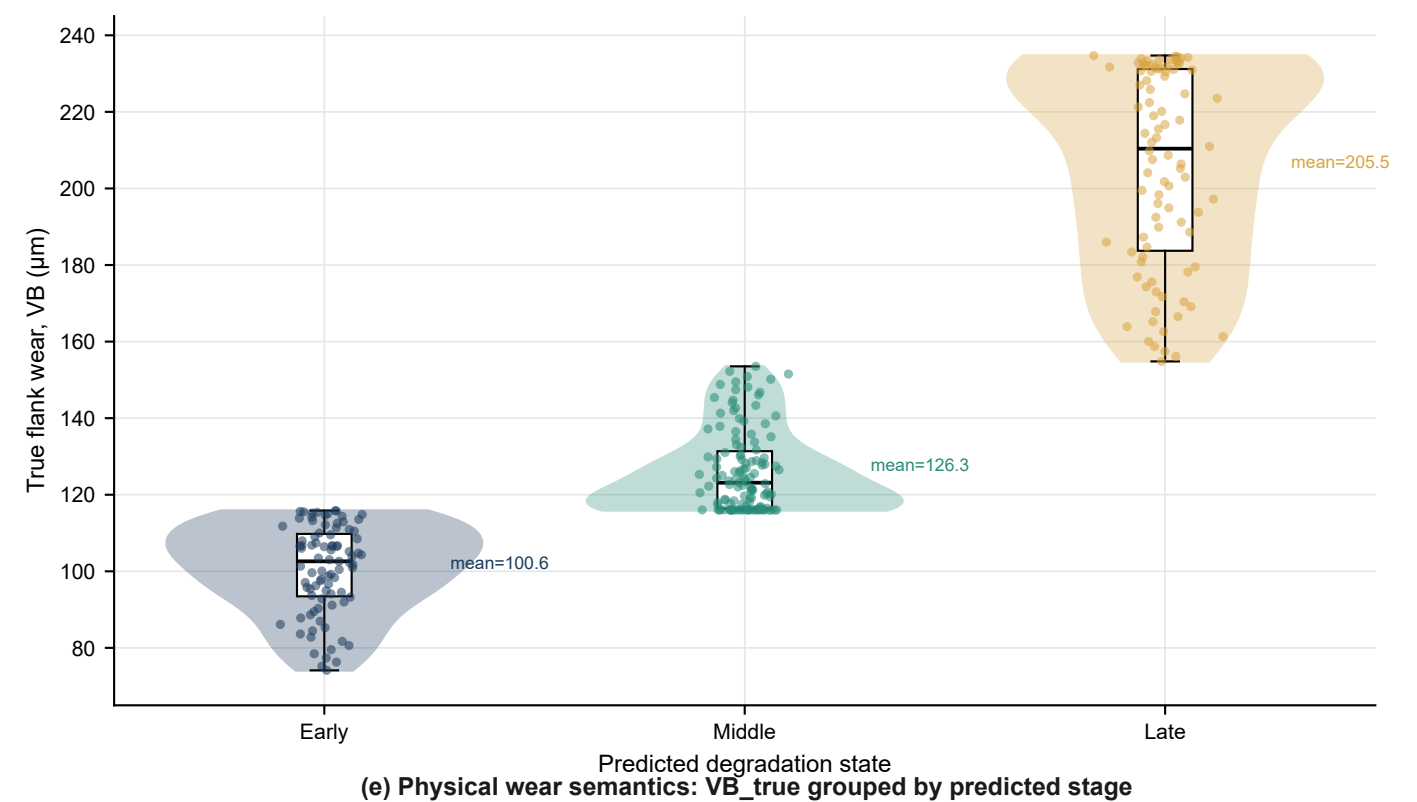
(b) Ordered trajectory in the probability simplex



(c) Continuous degradation-position agreement



(d) Shared latent manifold (64-D hidden representation, PCA; simple overview -- see Fig.5 for full analysis)



(e) Physical wear semantics: VB_{true} grouped by predicted stage